Emberz — speaker notes

SLIDE 1 · COVER

Emberz

“Everyone here has a parent or grandparent who picks up unknown calls. Emberz puts a guardian on that line.” Introduce yourselves in one line each.

SLIDE 2 · PROBLEM

The scam happens on the call. Nothing listens there.

Older Americans lost $7.7 billion to fraud last year — more than a third of all losses. When AI is involved, the average loss for someone over 60 triples, to $112,000. And it’s cheap: we built our own AI scam caller for 34 cents a call. Today’s filters judge the phone number before it rings. Once the call connects, nothing is listening. That’s where Emberz sits.

SLIDE 3 · HOW

Hear it, judge every turn, act.

The pipeline is simple: our number bridges the call, NVIDIA Nemotron Speech transcribes both sides live, and after every caller turn our guardian — fine-tuned on Fireworks — returns a verdict like this one, from the real call: urgent, fake bank call, a coach line for Margaret and a note for the family. It reads the family’s own rules. We trained it on generated calls, and more than a quarter of them are harmless calls that sound scary — a real bank alert, a pharmacy bot — so it learns when to stay quiet.

SLIDE 4 · HEADLINE

More calls handled right than Kimi K3.

We tested on 257 calls the model never saw, written by a different model family than our training data. Count every call: a scam caught before the ask, or a harmless call left alone. Our guardian handles 90.3%. Kimi K3, a frontier model, handles 83.7%. That gap is significant, and so is our gain over every other baseline. And when it catches a scam, it does so a median of four and a half turns before the caller asks for the code — that's the window where a whisper or a family member can still stop it.

SLIDE 5 · WINS

Quieter, and it follows the family's rules.

Where it really wins is the rest of the picture. It raises fewer false alarms — 18% of harmless calls, against 23% for Kimi K3 and 53% for the same model before fine-tuning. It follows the family's rules: same call, one rule changed, the right answer flips — we get 19 of 22 pairs, Kimi K3 gets 4. And on a public dataset it had never seen, it caught 98 of 100 scams with zero false alarms.

SLIDE 6 · VALUE

Small cost per family. Big pool of losses.

Why it's worth building: call-driven scams — tech support, government impostors, romance — cost $3.86 billion a year, so every 1% we stop is worth $38.6 million. On our cost model it runs for about $3.64 per family per month, line and model included. The buyer is the adult child with an aging parent; today's tools either warn only the person being manipulated or stop at the ring. Emberz is the one that brings in the family.

SLIDE 7 · DEMO

Our AI scammer calls Grandma. Watch.

Hand over to the demo: “Enough slides. We’re going to have our AI scammer call this phone right now.” Run: bash demo/run_demo.sh bank_fraud (it preflights, pauses batch jobs and places one AI call). Put the protected phone on speaker and show the caretaker console. Point out the four moments on this slide as they happen. If the line fails, use the backup slide with the real 15:48 call or play demo/video/renders/emberz_demo_final.mp4. Tip: run the guardian through the sampler proxy (the exact evaluated checkpoint), as on the 15:48 call; the dedicated deployment follows family rules less well. After the demo, come back to the closing slide.

SLIDE 8 · CLOSE

Give Grandma a guarded number.

“Scammers now have AI for 34 cents a call. Emberz puts a guardian on the line — measured against a frontier model, and working on real phones today. Give Grandma a guarded number. Thank you.” Backup slides for questions: the 19-slide deck (ablations, competitors, consent, cost details).

SLIDE 9 · CALL

An AI scammer calls “Margaret”. 94 seconds.

Here’s a real call from yesterday. Our AI scammer, “Daniel from First Meridian”, calls Margaret. Five seconds in, our detector flags the voice as synthetic. When he claims a suspicious $842 charge, the guardian raises an urgent alert, whispers to Margaret only — “Don’t confirm anything, hang up and call the number on your card” — and phones the family. The caretaker joins and listens. When the caller says he’s sending a verification code, a second alert fires, and the caretaker presses 9. The scammer is dropped. Ninety-four seconds, no code shared. [Optional: play the 2-minute video here.]

SLIDE 10 · NEXT

What we haven’t solved yet.

We want to be straight about what’s left. The deployed model follows family rules less well than the checkpoint we benchmarked — we think the adapter isn’t fully applied in serving, and we’ve raised it with Fireworks. The benchmark calls are generated, so real role-played calls are next. And one-way robocalls aren’t what this is for. On privacy: every call starts with a notice, the protected adult opts in themselves, and nothing is kept past seven days.